

Relational Geometry: The Minimal Economy of Uncertainty

Ontological foundation and derivational origin of a parallel framework for physical understanding

Le Matt Ansatz Di Ego

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1 Abstract

The aim of this proposal is to present a minimal ontological economy of known reality and to provide a conceptual foundation for a parallel framework of physical understanding.

The proposal is not intended to replace quantum mechanics, relativity, or quantum field theory. It studies the same reality from a different starting point. Instead of beginning with objects located in space, states evolving in time, or forces acting between bodies, it asks what minimal relation allows a difference to be preserved until it constitutes an identity.

The framework begins with Postulate 0: totality admits a non-trivial self-comparison. This comparison does not divide totality into independent objects, but it allows a first distinguishable expression. A subsequent comparison opens a difference. The difference imposes preservation conditions that may receive a formal representation. A solution is not reduced to a value: it jointly comprises a configuration and the domain required to preserve it. That pair constitutes uncertainty in the specific sense of the program.

Space appears as the domain of a solution. Unique information consists of the invariants that must be preserved for the solution to remain the same. Identity is a derived class of configurations preserving those invariants. Relational Geometry is the minimally sufficient architecture that preserves the solution, its domain, and its unique information.

Relational economy does not require closing all openness. It requires preserving the constitutive openness of identity, resolving only what has sufficient conditions, and introducing no unnecessary relations. At the abstract level, this economy is expressed by a partial order among sufficient architectures. In concrete realizations it may acquire a structural measure, such as the rank of a closure system.

Version 2.1.0 explicitly formulates self-comparison, removes unjustified norms from the ontological level, distinguishes information from identity, conditions inter-system distance on the existence of a shared metric, generalizes quadratic closure through an interaction term, and positions the program relative to other relational traditions. A companion document, *A First Finite Closure Realization of Relational Geometry*, builds a computable toy model and executes several internal falsifiers for the first time.

No experimental validation or demonstrated equivalence with established physical theories is claimed.

Keywords: relational geometry; constitutive uncertainty; spatial solution; identity; unique information; closure operator; minimal economy; rank; emergent dimension; orthogonal closure.

2 Editorial note

August 14, 2026 is the planned publication date. This revision was prepared on July 29, 2026.

The short declarative register of some sections functions as axiomatic notation. Sentences such as “totality simply is” do not replace demonstration and are not intended as mystical language. They delimit the level at which the construction begins.

3 1. Purpose and position of the program

Contemporary physics describes observable regularities with extraordinary precision. It calculates energies, probabilities, trajectories, fields, curvatures, and transitions.

Relational Geometry asks a complementary question:

What is the minimal relational expression from which those categories could emerge?

The program is conceived as a **parallel framework of understanding**.

Parallel means that it:

- studies the same physical reality;
- seeks to preserve established empirical results;
- does not need to deny equations that already work;
- changes the conceptual starting point;
- attempts to derive categories that other descriptions treat as primitive.

Its success does not depend on using different language. It depends on reconstructing known results through a more fundamental structure and, ultimately, producing a distinctive consequence that can be falsified.

4 2. Central thesis

The thesis may be written as one sequence:

difference $\longrightarrow$ preservation conditions $\longrightarrow$ solution and domain $\longrightarrow$ geometry $\longrightarrow$ unique information $\longrightarrow$ identity.

In words:

A difference is preserved as a relational solution. The extension required to preserve it constitutes the space it occupies. The minimal organization of that extension constitutes its geometry.

The guiding principle follows:

Relational Geometry is the minimal economy required to preserve the uncertainty that constitutes an identity.

5 3. Epistemic discipline

Every claim must belong to an explicit level.

5.1 3.1 Ontological postulate

An initial condition adopted to open the framework.

5.2 3.2 Structural definition

A term introduced to express a previously distinguished relation.

5.3 3.3 Internal formal result

A consequence obtained within an explicit mathematical realization.

5.4 3.4 Operational bridge

A rule connecting an internal magnitude with a potentially measurable quantity.

5.5 3.5 Tentative physical correspondence

An interpretation relating a structure of the program to a physical phenomenon without presenting it as a derivation.

5.6 3.6 Falsifiable hypothesis

A quantitative statement with success and failure conditions fixed before data evaluation.

5.7 3.7 Programmatic speculation

A research direction that cannot be used as evidence for the previous layers.

5.8 3.8 Status of the falsifiers

In the abstract document, falsifiers are conceptual protocols. They are not all executable until the mathematical classes of the structures involved are specified.

The finite companion model executes the domain, economy, sufficiency, information-identity, and quadratic-closure falsifiers inside a declared realization. It does not execute the emergence of the mathematical ambient class: it explicitly uses a finite vector space as toy scaffolding.

6 4. Postulate 0: non-trivial self-comparison

Postulate 0. Totality admits a relation with itself capable of producing a distinguishable expression while preserving totality as reference.

This is the fundamental generative step. The proposal does not claim that a structureless totality spontaneously produces structure through an already known operation. It postulates that self-comparison may be non-identical.

$$\mathcal{T} \xRightarrow{\text{non-trivial self-comparison}} \mathbf{U}.$$

Every later realization must state how it represents this postulate and what restrictions it places on the first expression.

7 5. Totality and notation discipline

7.1 5.1 Totality

Totality simply is.

It is represented by:

$$\mathcal{T} \equiv \hat{\aleph}.$$

Complete reference is marked by:

$$N(\mathcal{T}) = 1.$$

Here N is not a vector norm. It is a referential normalization marker. The symbol 1 does not denote length, energy, or physical cardinality.

At this level there is not yet a need for objects, systems, particular identities, distances, dimensions, differential information, space, or time.

Totality is not a system, because a system requires organized differences.

7.2 5.2 No hidden mathematical structure

Norms, inner products, and metrics will not be used at the ontological level unless explicitly declared mnemonic.

The former notation $\|\hat{\mathbb{N}}\| = 1$ is replaced by $N(\mathcal{T}) = 1$. A radius magnitude is introduced only when a realization defines a scale function.

8 6. First relation and first radius

Self-comparison acquires a scale structure represented by:

R.

Definition 6.1 - First radius. The first radius is the minimal scale relation through which the self-comparison of totality acquires a first distinguishable expression.

The word *scale* does not yet presuppose a physical unit. It means that expressions may be compared through a correspondence relation.

In a realization equipped with a scale function s , one may write:

$$r = s(\mathbf{R}).$$

The historical algebraic program writes:

$$\mathbf{U} = J_{\mathbf{R}}(\hat{\mathbb{N}}).$$

The operator remains to be constructed. The core preserves only the dependence:

$$\mathcal{T} \xRightarrow{\mathbf{R}} \mathbf{U}.$$

9 7. To exist and to be existing

To exist is to acquire a first relational expression without yet requiring a domain of identity of one's own.

To be existing is to preserve an identity through a solution that determines and occupies a relational domain of its own.

The distinction is logical:

to exist $\longrightarrow$ primary expression,

to be existing $\longrightarrow$ solution, domain, and identity.

10 8. Difference and preservation conditions

A first expression is compared again. The new comparison may fail to contain the previous relation completely.

Let δ denote a relational difference. It imposes preservation conditions:

$$\mathcal{C}_\delta.$$

An equation is not the ontological cause of the difference. It represents those conditions:

$$\mathcal{C}_\delta \mapsto \mathcal{E}_\delta[\Psi, \Omega] = 0.$$

This separation preserves the priority of relation over mathematical language.

11 9. Uncertainty as a spatial solution

11.1 9.1 Terminological note

Uncertainty here denotes a relational solution together with the domain and constitutive openness preserving its identity.

It is not automatically identical to epistemic uncertainty, experimental error, statistical entropy, quantum superposition, or Heisenberg uncertainty. Any such connection must be derived.

11.2 9.2 Ambient class

An equation with variable domain requires a declared ambient class of candidates.

Let $\mathfrak{D}_\delta$ be a typed category, class, or collection of admissible pairs (Ψ, Ω) . Each realization must specify:

- admissible domains;
- configurations over those domains;
- admissible transformations;
- equivalence between domains.

The presence of $\mathfrak{D}_\delta$ is explicit. The abstract program does not claim to have derived that class.

11.3 9.3 Joint solution and domain

$$(\Psi_\delta, \Omega_\delta) \in \text{Sol}_{\mathfrak{D}_\delta}(\mathcal{E}_\delta),$$

and:

$$\mathfrak{U}_\delta := (\Psi_\delta, \Omega_\delta).$$

Definition 9.1 - Uncertainty. Uncertainty is a relational solution jointly constituted by a configuration and the domain required to preserve it.

This notation declares co-determination but does not prove it. A realization must construct a rule linking configuration and domain. The finite companion model uses $B \mapsto \text{cl}(B)$.

11.4 9.4 Constitutive and descriptive openness

$$U_c = \text{constitutive uncertainty}, \quad U_d = \text{descriptive uncertainty}.$$

Constitutive uncertainty delimits identity. Descriptive uncertainty is avoidable structure of a representation. Relational economy seeks to reduce U_d without destroying U_c .

12 10. Latent solutions and domain equivalence

Let $\cong_\delta$ be an equivalence preserving the relevant invariants. The latent class of $\mathfrak{U}_\delta$ is:

$$\Sigma(\mathfrak{U}_\delta) = \left\{ (\psi, \Omega) \in \text{Sol}_{\mathfrak{D}_\delta}(\mathcal{E}_\delta) : (\psi, \Omega) \cong_\delta (\Psi_\delta, \Omega_\delta) \right\}.$$

A realization may require domains to be equal, isomorphic, or equivalent under a specified transformation. In the finite closure model, latent configurations are minimal bases generating the same closed domain:

$$\text{Bas}(\Omega) = \{B : \text{cl}(B) = \Omega, B \text{ minimal}\}.$$

13 11. Space as the domain of a solution

At the structural level:

$$\text{proper space of the solution} \equiv \Omega_\delta.$$

Principle 11.1 - Spatial emergence. The space of a solution is the relational extension required to preserve its difference.

This does not yet derive shared physical space. A complete theory must explain how distinct domains are compared and how a common metric emerges.

14 12. Relational Geometry

Let $\mathcal{R}_\delta$ be the family of relations organizing a solution in its domain. Define:

$$\mathcal{G}_\delta = (\Omega_\delta, \mathcal{R}_\delta).$$

Definition 12.1 - Relational Geometry. Relational Geometry is the architecture of extension and relations through which a solution preserves its difference.

Geometry does not presuppose a Euclidean metric.

15 13. Relational economy

15.1 13.1 Canonical partial order

$$\mathcal{G}_1 \preceq_E \mathcal{G}_2$$

means that $\mathcal{G}_1$:

- preserves at least the same relevant invariants;
- adds no more independent relations than necessary;
- introduces no closure without sufficient conditions;
- retains no descriptive openness that its architecture can already resolve.

Architectures may remain incomparable.

15.2 13.2 Conditional scalar measure

A function $K(\mathcal{G})$ may be used only when a realization defines a structural measure invariant under its equivalence relation.

Then:

$$\mathcal{G}_\delta^* \in \arg \min_{\mathcal{G} \in \mathfrak{A}(\mathfrak{U}_\delta)} K(\mathcal{G}),$$

subject to:

$$\text{Inv}_{\mathcal{G}}(\mathfrak{U}_\delta) = \mathcal{J}_\delta.$$

In a finite matroid, rank provides such a measure. Outside that realization, the partial order has priority.

16 14. Unique information and identity

Definition 14.1 - Unique information. Unique information is the set of relational invariants that must be preserved for a solution to remain the same.

$$\mathcal{J}_\delta = \text{Inv}(\Psi_\delta, \Omega_\delta, \mathcal{R}_\delta).$$

Let G_δ be a family or group of admissible transformations preserving $\mathcal{J}_\delta$. Identity is the orbit:

$$\mathcal{J}_\delta = [\Psi_\delta, \Omega_\delta, \mathcal{R}_\delta]_{G_\delta}.$$

Definition 14.2 - Identity. Identity is the class of configurations and domains preserving the same unique information under admissible transformations.

The asymmetry is explicit:

$$\mathcal{J}_\delta \longrightarrow \mathcal{J}_\delta.$$

17 15. Radius and distance between identities

The following construction is conditional on a shared phase space, a distance d , and defined identity radii.

$$I_i = (\phi_i, r_i), \quad \mathcal{B}_i = \{\phi : d(\phi, \phi_i) \leq r_i\}.$$

Conditional Definition 15.1 - Identity radius. In a metric realization, identity radius delimits variation preserving the invariants of an identity.

$$D_{ij} = \frac{d(\phi_i, \phi_j)}{r_i + r_j}.$$

The interpretation $D_{ij} < 1$, $= 1$, or > 1 describes overlap, boundary, and separation only within that realization.

Pairwise overlap need not be transitive. A latent class therefore requires an additional rule such as connected components, transitive closure, or a stronger equivalence.

18 16. System and hierarchy

$$\mathfrak{S} = (\mathfrak{U}, \mathcal{G}, \mathcal{I}).$$

Definition 16.1 - System. A system is an architecture of solutions whose geometry preserves a distinguishable identity.

A stabilized identity at one level may function as an open solution at another:

$$\text{solutions} \longrightarrow \text{identities} \longrightarrow \text{systems} \longrightarrow \text{new comparisons}.$$

19 17. Novelty, sufficiency, and dimension

$$\nu_{n+1} \notin \text{Cl}(\mathcal{R}_n).$$

Definition 17.1 - Novelty. Novelty is an independent relation whose preservation requires enlargement of the architecture.

While $\text{Suf}(\mathcal{G}_n, \mathfrak{U}) = 1$, no enlargement is required. When it becomes zero, the minimal independent relation restoring sufficiency is added:

$$\mathcal{G}_{n+1} = \mathcal{G}_n \oplus A_{\min}.$$

Hypothesis 17.2 - Dimensional emergence. A dimension emerges when an independent difference requires a capacity for distinguishability that cannot be represented within the previous architecture.

In the matroid realization, this becomes a theorem: an independent relation increases rank by one.

20 18. Resolution

Resolution may consist of closure, stable containment, class separation, transfer to another level, formation of a composite identity, or explicit preservation of constitutive latency.

Principle 18.1 - Sufficient resolution. A resolution is legitimate when it reduces only the openness for which sufficient conditions exist and preserves the openness still belonging to the solution.

21 19. Quadratic closure and interaction

If a realization has a bilinear form and:

$$I = V + m,$$

then:

$$Q_I = Q_V + Q_m + 2Q_{\times}, \quad Q_{\times} = \langle V, m \rangle.$$

Strict additivity:

$$Q_I = Q_V + Q_m$$

holds when:

$$\langle V, m \rangle = 0.$$

Hypothesis 19.1 - Orthogonal closure. The expressed channel and the complement separate quadratically when they share no relational component under the adopted bilinear form.

The cross term directly falsifies exact binary closure.

The binary depth is a modeling choice:

$$q_\varepsilon = 2^{-\varepsilon}, \quad \varepsilon \in \mathbb{N}_{\geq 1},$$

with $\varepsilon = 0$ reserved as the no-complement reference case.

If $Q_k \leftrightarrow \omega_k^2$ and $\mathcal{R}_k = c/\omega_k$, orthogonal closure implies:

$$\frac{1}{\mathcal{R}_I^2} = \frac{1}{\mathcal{R}_V^2} + \frac{1}{\mathcal{R}_m^2},$$

hence:

$$\mathcal{R}_I < \min(\mathcal{R}_V, \mathcal{R}_m).$$

This operational compression depends on prior definitions of frequency, time, and causal scale.

22 20. Octave relation

$$\vartheta(x) = \log_2 \left(\frac{x}{x_0} \right) \bmod 1, \quad \vartheta(x) = \vartheta(2^n x).$$

This is mathematically the form of discrete scale invariance. A possible physical signature would be log-periodic modulation, but no such correspondence is claimed here.

23 21. Tentative physical bridges

The program investigates, without claiming derivation:

$$\mathbf{U} \overset{?}{\longleftrightarrow} \text{light as first relational expression,}$$

$$\text{photon} \overset{?}{\longleftrightarrow} \text{discrete shared luminous difference,}$$

$$\mathfrak{P} = (\Psi_P, \Omega_P, \mathcal{R}_P, \mathcal{J}_P)$$

as a representation of a stabilized particle.

Bridges using ω , c , or physical distance depend on prior constructions of time, metric, causality, and operational units.

24 22. Time, shared space, and force

Time is investigated as an ordered comparison between identity transformations:

$$\mathcal{J}_n \longrightarrow \mathcal{J}_{n+1}.$$

Shared space requires compatibility rules between domains:

$$\Omega_A \leftrightarrow \Omega_B.$$

Force is investigated as the dynamical manifestation of readjustment between incompatible relational closures. None of these constructions is complete.

25 23. Related programs

Barbour and Bertotti formulated a Machian dynamics relational in motion and time [2]. Rovelli's relational quantum mechanics treats states and physical values as relative to interactions [3]. Causal set theory proposes a locally finite partial order underlying spacetime [4]. Ontic structural realism gives ontological priority to structure [5]. Constructor theory formulates laws in terms of possible and impossible transformations [6]. Matroid theory abstracts dependence, independence, bases, and rank [1]. Discrete scale invariance connects preferred scale ratios with log-periodic signatures [7].

Relational Geometry differs through the combined proposal of:

1. joint determination of solution and domain;
2. constitutive uncertainty as identity boundary;
3. unique information prior to identity;
4. relational economy as a selection principle;
5. dimension as minimal enlargement of distinguishability;
6. closure as stabilization rather than origin of identity.

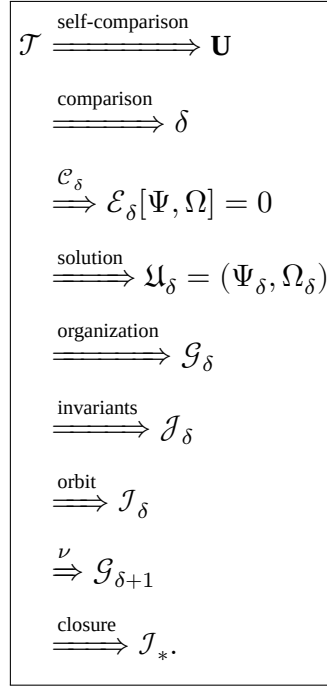
26 24. Historical algebraic program

The historical sequence is retained:

$$\hat{\mathbb{N}} \longrightarrow \mathbf{R} \longrightarrow \mathbf{U} \longrightarrow \mathbf{V} \longrightarrow \mathbf{m}.$$

The historical identity $\mathbf{U} = \mathbf{V} + \mathbf{m}$ is not used as a theorem. Closure is expressed through quadratic quantities and an interaction term until a consistent algebraic realization is available.

27 25. Canonical derivational sequence



28 26. Internal falsifiers

1. **Domain:** if one configuration admits arbitrary domains without equivalence or selection, co-determination has not been constructed.
2. **Prior ambient structure:** if a realization claims to derive space while using a complete physical metric as input, the emergence is circular.
3. **Economy:** if an independent relation can be removed without changing $\mathcal{J}$, the architecture is not minimal.
4. **Sufficiency:** if the minimal architecture loses a relevant invariant, reduction has gone too far.
5. **Information and identity:** an invariant set fails if it cannot separate the intended identity classes.
6. **Closure:** exact quadratic closure fails when $Q_\times \neq 0$.
7. **Physical bridge:** if a correspondence merely renames known equations without new constraints, the program remains a reinterpretation.

29 27. First execution of the falsifiers

The companion document uses the Fano geometry, the seven non-zero vectors of $GF(2)^3$.

Protocol	Finite-model result
Closure axioms	Passed
Basis exchange	Passed
Generated domain	Unique inside the model
Economy as minimal basis	Passed
Sufficiency	Passed
Information prior to identity	Constructed
Novelty increases rank	Passed

Protocol	Finite-model result
Orthogonal closure	Discriminated by cross term
Emergence of the ambient class	Not demonstrated

The negative final result is deliberate. The toy model declares $GF(2)^3$ as its ambient structure and tests the architecture downstream of differentiation, not the ontological origin of that ambient structure.

30 28. Conditions for physical distinguishability

The program must produce a consistent mathematical realization, operational definitions, parameters fixed without retrospective fitting, comparison with baseline models, a distinctive quantitative prediction, and a failure criterion declared in advance.

Compatibility with known results is necessary but not sufficient.

31 29. Priority open problems

1. Construct a realization of Postulate 0.
2. Specify a minimal ambient class of domains without importing complete physical space.
3. Generalize beyond representable matroids.
4. Define physically meaningful identity invariants.
5. Derive identity radius from those invariants.
6. Construct a shared metric between domains.
7. Derive a genuine uncertainty inequality limiting simultaneous resolution of paired openings.
8. Derive or falsify orthogonal closure.
9. Justify or replace the binary choice.
10. Derive time, causality, and dynamics.
11. Construct electromagnetic, material, and gravitational bridges.
12. Produce a physically distinctive prediction.

32 30. Conclusion

The proposal begins with a non-trivial self-comparison of totality. A subsequent comparison opens a difference. The difference imposes preservation conditions. A solution of those conditions comprises a configuration and the domain required to preserve it. That pair constitutes uncertainty.

The domain is the proper space of the solution. Its organization is geometry. Its invariants are unique information. The class of configurations preserving those invariants is identity. Relational Geometry selects, through an economy order, the minimally sufficient architecture preserving that identity.

The finite companion document shows that this chain can receive a first computable realization. Closure, basis, rank, novelty, dimension, information, identity, and orthogonal closure become executable. The model does not yet derive totality or physical space; it declares precisely where its formalization begins.

The central formulation remains:

Uncertainty is the solution that defines the space of an identity; Relational Geometry is the minimal economy required to preserve it.

Signed: Le Matt Ansatz Di Ego

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